

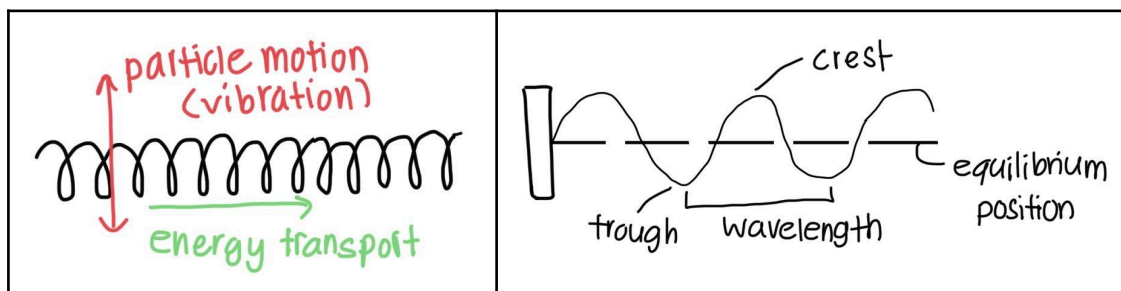
T3: Waves (3.1 - 3.2)

3.1 Properties of waves

- Waves transfer energy without transferring matter

Transverse waves:

- The direction of vibration is perpendicular to the direction of propagation (or energy transfer/wave travel)



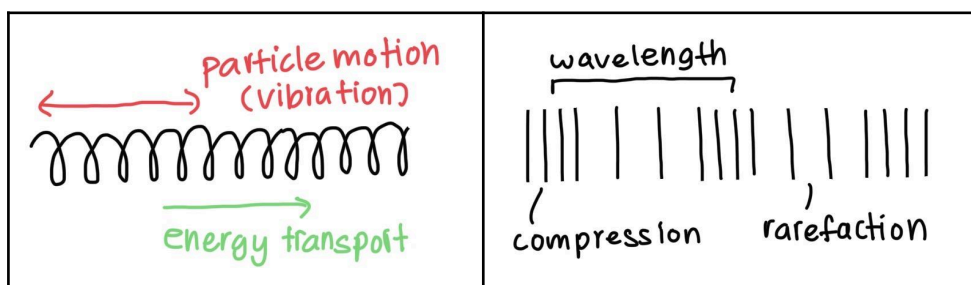
Examples - water waves, seismic secondary waves, electromagnetic waves (including light waves)

Crest: the highest point on a wave, above its equilibrium position

Trough: the lowest point on a wave, below its equilibrium position

Longitudinal waves:

- The direction of vibration is parallel to the direction of propagation (or direction of energy transfer/wave travel)



Examples: Sound waves, seismic primary waves

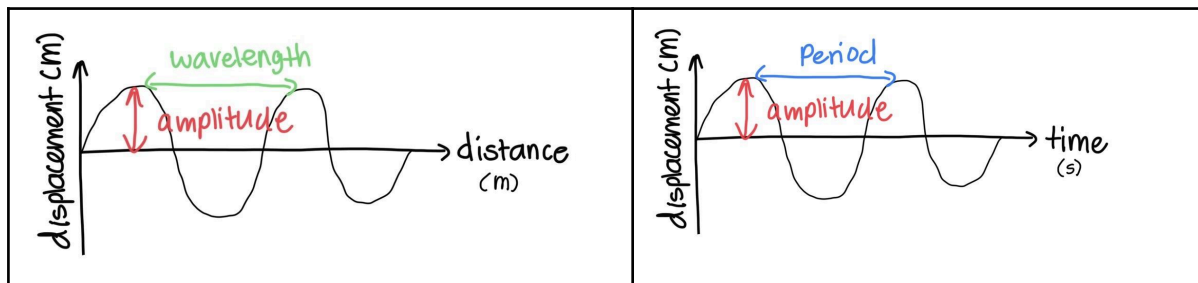
Compression: region where wave particles are close together

- high pressure and high density

Rarefaction: region where wave particles are far apart

- low pressure and low density

Describing waves



Amplitude: the distance from equilibrium position to the peak of a wave

Wavelength: the distance between 2 consecutive peaks

Frequency: the number of vibrations / waves that travel past a point, per second

Period: the time taken for 1 vibration / 1 wave to travel past a point

$$\text{frequency} = \frac{1}{\text{period}}$$

Wave speed: the distance travelled by the wavefront per unit time

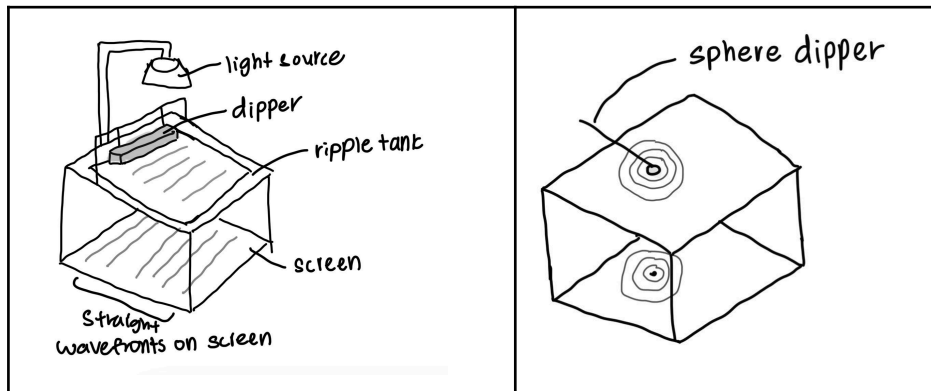
$$v = f \lambda$$

Labels in the diagram:
- v is labeled 'wave speed (m/s)' with a curved arrow pointing to it.
- f is labeled 'frequency (Hz)' with a curved arrow pointing to it.
- λ is labeled 'wavelength (m)' with a curved arrow pointing to it.

Wavefront: a line joining adjacent points on a wave that are all in phase

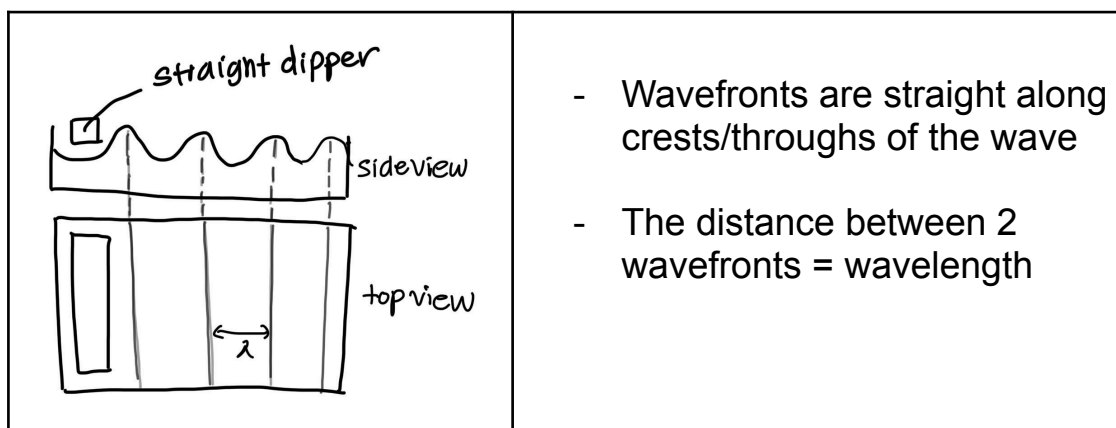
Ripple tank and wavefronts

Ripple tank:



- Light illuminates wavefronts, making them look visible
 - Screen below ripple tank is used to observe wavefronts of the water wave
-
- A straight dipper is used to create straight wavefronts
 - A sphere dipper is used to create circular wavefronts

Wavefronts:

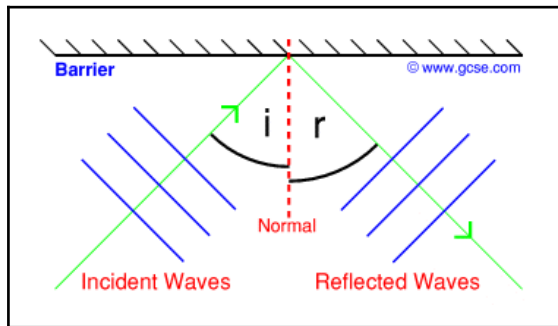


- Wavefronts are always perpendicular to direction of propagation (energy transfer/wave travel), regardless whether transverse or longitudinal

Reflection of waves

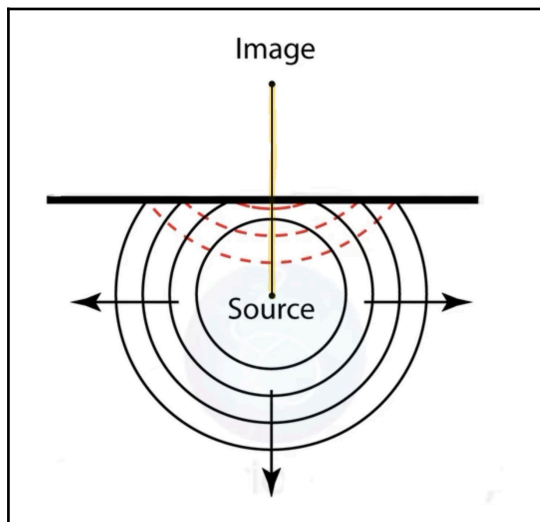
- When waves hit an obstacle, they are reflected
- The direction of wave propagation changes
- Speed, wavelength and frequency remain constant

For straight wavefronts:



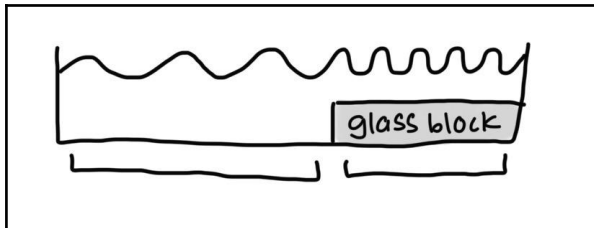
For circular wavefronts

- Both source and image are same distance from obstacle



Refraction of waves

- When waves travel from one medium to another, they are refracted
- This is because the speed of the wave changes, which causes wavelength to also change.
- Frequency remains constant



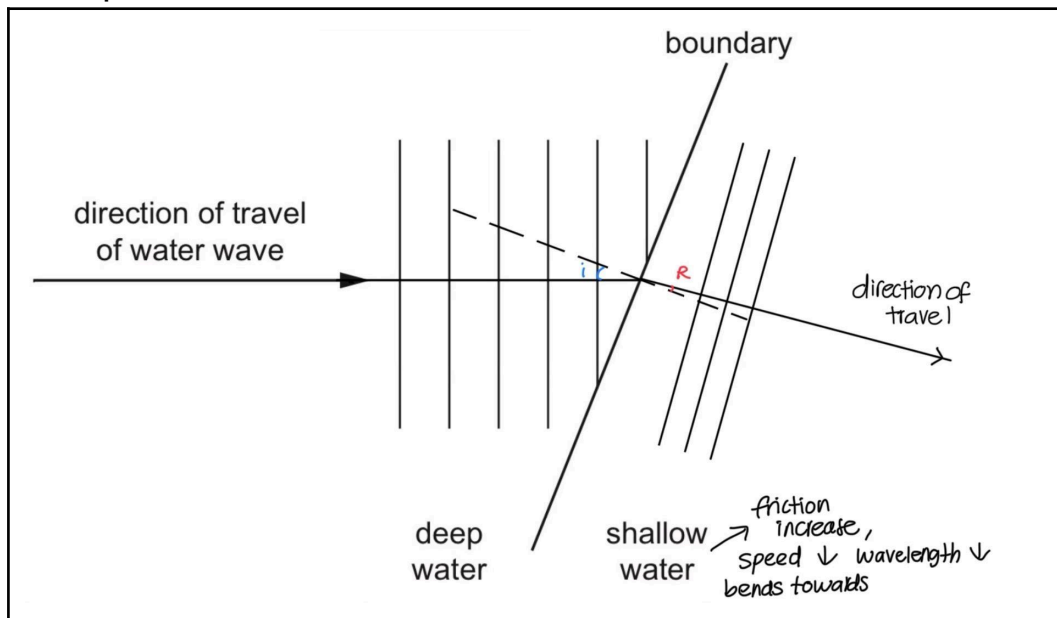
In deep water,

- Friction decreases
- Speed increases
- Wavelength increases

In shallow water,

- Friction increases
- Speed decreases
- Wavelength decreases

Example:



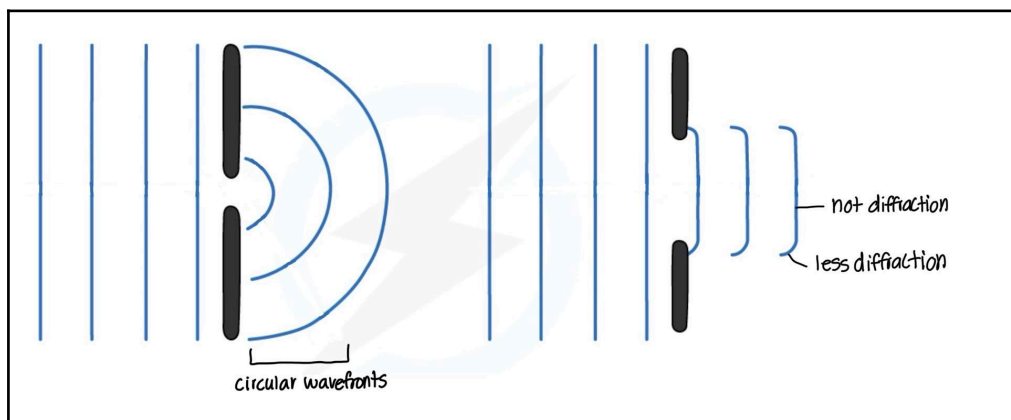
- Since travelling from deep to shallow water, friction increases, speed decreases and wavelength also decreases
- Since slower, bends towards normal

Diffraction of waves

- The spreading out of waves as they pass through a gap, or around an edge
- After diffraction, speed, wavelength and frequency remains the same

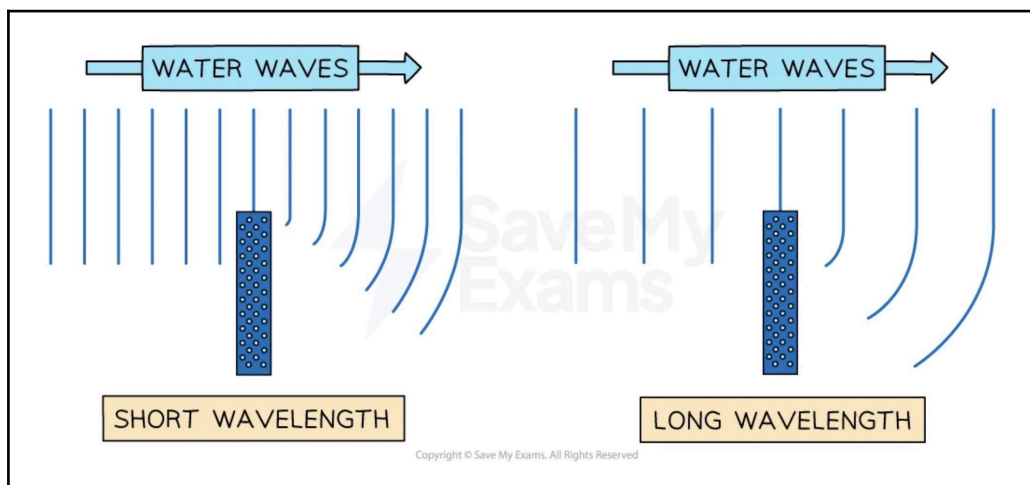
Diffraction through a gap:

- If gap is similar or smaller than wavelength, there is more diffraction
- If gap is larger than wavelength, there is less diffraction



Diffraction around an edge:

- If wavelength is long, more diffraction
- If wavelength is short, less diffraction



- It is like straight, then curved at the edges of the wavefront

3.2 Light

Properties of light:

- Light waves are electromagnetic waves, so they can transfer energy through a vacuum
- Speed of light in vacuum is $3 \times 10^8 \text{ m/s}$, which is approx. equal to speed of light in air
- Light travels in a straight line
- Light waves reflect, refract and diffract

Reflection of light

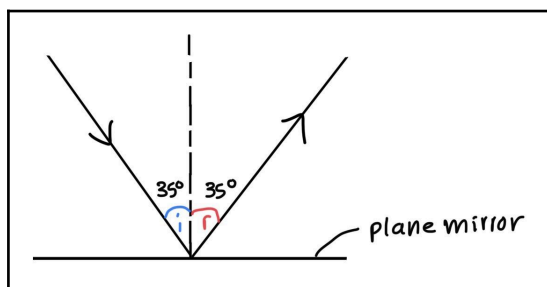
- Angle of incidence (i) = angle of reflection (r)
- Incident ray, reflective ray and normal lie on the same plane

Normal - the line drawn 90° to a surface, at the point where a ray hits the surface

Angle of incidence - the angle between the incident ray and the normal

Angle of reflection - the angle between the reflected ray and the normal

Normal reflection:



Formation of image in a mirror:

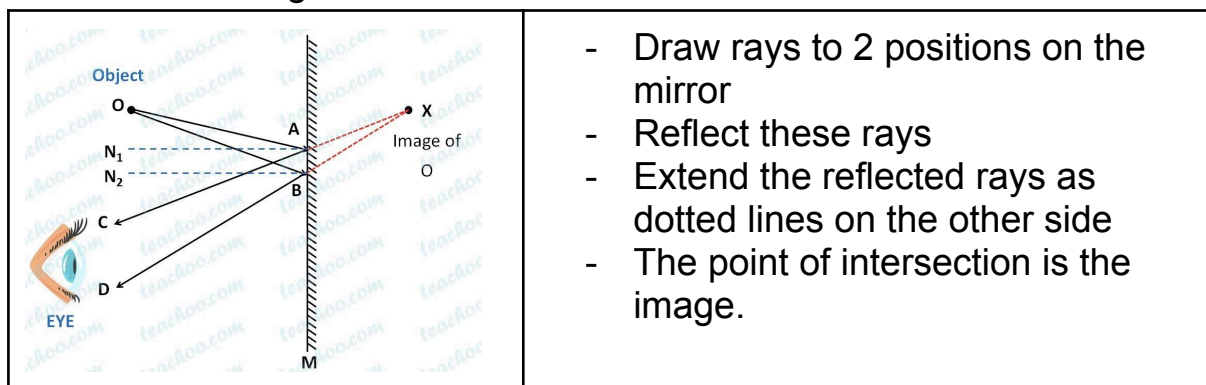


Image characteristics:

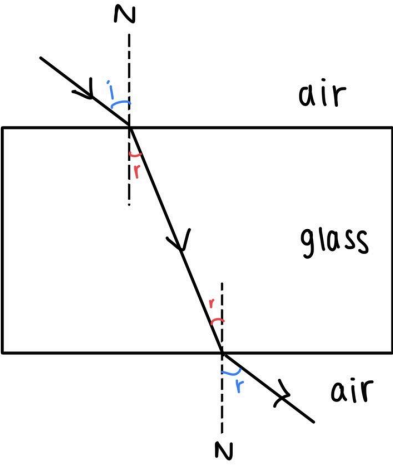
- Virtual - cannot be formed on screen
 - Same size as object
 - Same distance from plane mirror
 - Laterally inverted
-

Refraction of light

- Light will refract when it travels through a transparent material, such as glass
- Again: wavelength and speed changes, frequency remains the same

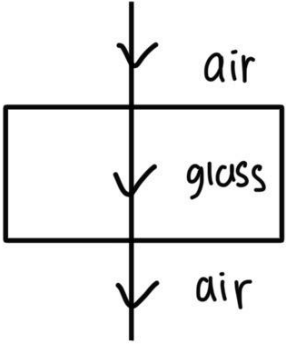
Angle of refraction - the angle between the refracted ray and the normal

Entering and leaving a glass block:

	- When light ray travels to a less dense medium, it speeds up - When light ray travels to a more dense medium, it slows down - As the light ray enters the glass block, it slows down, and bends towards the normal - As the light ray leaves the glass block, it speeds up, and bends away from the normal
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- Same -> Speed increases meaning wavelength increases, & speed decreases meaning wavelength decreases

When angle of incidence = 0° :

	- When angle of incidence = 0°, the ray enters along the normal, 90° to the surface of the glass block - When it enters the glass block, the light slows down but does not change direction. the angle of refraction also = 0° - When it leaves the glass block, the light speeds up but does not change direction
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- Also $\rightarrow$ air > water > glass (for speed of light)

Refractive index

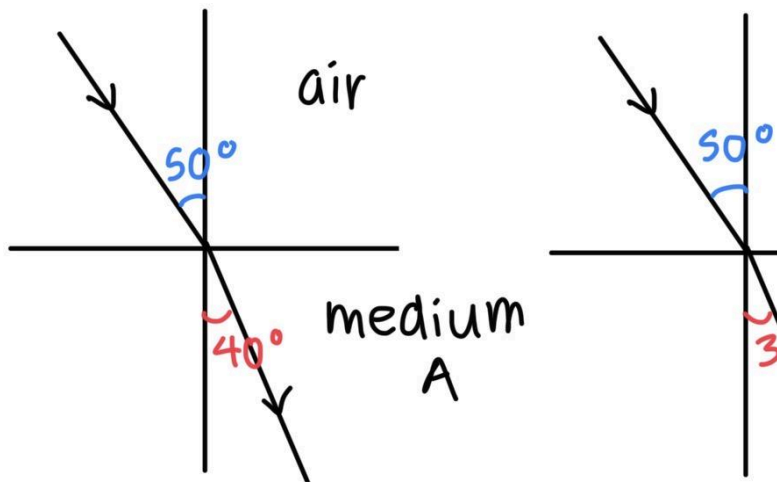
Refractive index: the ratio of speed of light in a vacuum to speed of light in a medium

- The refractive index of a material is related to how dense it is
- The denser the material, the higher the refractive index

$n = \frac{c}{v}$ n — refractive index c — speed of light in vacuum (m/s) v — speed of light in medium
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- The Refractive index of air is 1

Example:



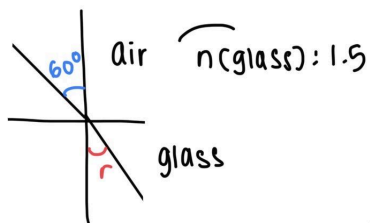
Medium B bends more towards the normal than medium A

- Speed (medium A) > speed (medium B)
- Density (medium A) < density (medium B)
- Refractive index (medium A) < refractive index (medium B)

Refractive index using angles: (Snell's law)

$$n_1 \sin i = n_2 \sin r$$

Example 1:

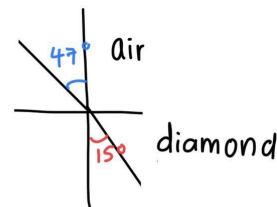


$$n_1 \sin i = n_2 \sin r \quad r = \sin^{-1} \left(\frac{\sin 60}{1.5} \right)$$

$$(1) \sin 60 = 1.5 \sin r = 35.3^\circ \checkmark$$

$$\sin r = \frac{\sin 60}{1.5}$$

Example 2:

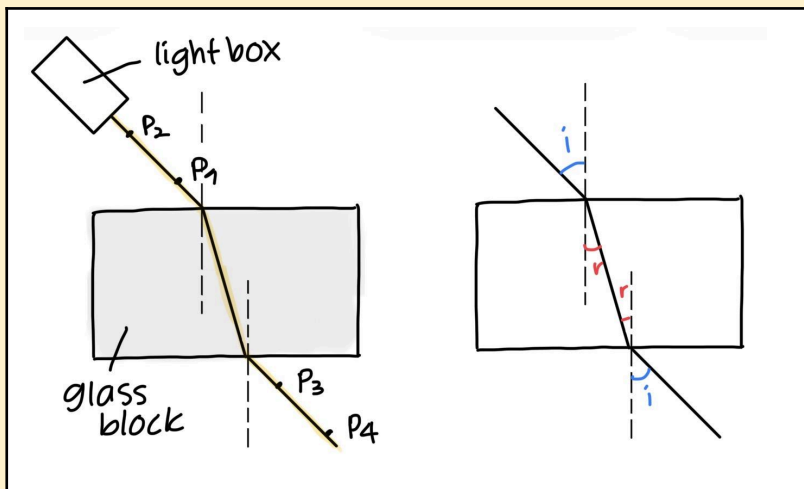


$$n_1 \sin i = n_2 \sin r$$

$$(1) \sin 47 = n_2 \sin 15$$

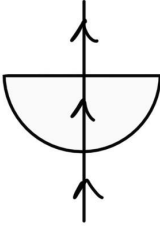
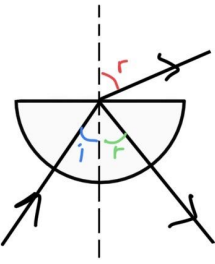
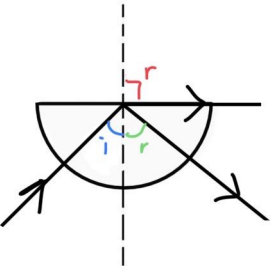
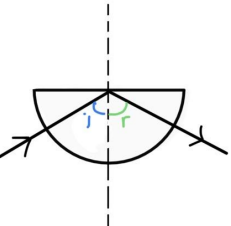
$$n_2 = \frac{\sin 47}{\sin 15} = 2.8 \checkmark$$

An experiment to investigate the refraction of light, and calculate the speed of light in glass or refractive index



- Place a glass block on top of a piece of paper, and trace around the block using a pencil
- Using a light box, shine a ray of light 30° to the glass block
- Place 2 pins along the ray entering the glass block
- Place 2 pins along the ray emerging from the glass block
- Remove the glass block and pins, and mark at the pin holes
- Turn off the light box.
- Use a pencil and ruler to draw and connect the pins, and extend them to draw the incident ray and the emerging ray.
- Then, connect the end of the incident ray and the start of the emerging ray to show the path of light in the glass block
- Draw normal lines, and measure angle of incidence (i) and angle of refraction (r)
- Calculate refractive index using $n = \sin(i) / \sin(r)$
- Calculate speed of light in medium by using $v = c/n$ (from $n = c/v$)
- Repeat for angles of 40, 50, 60, and 70°.
- Plot graph of $\sin i$ against $\sin r$. Gradient = refractive index.

Critical angle and total internal reflection

	- Light does not change direction, because angle of incidence = 0°, ray enters along normal, 90° to glass block - So, angle of refraction also = 0°
	- When angle of incidence, some light reflects and some light refracts - Refraction is from more dense to less dense medium, so speed increases, bends away from normal (refraction angle > incidence angle)
	- When angle of incidence increases, angle of refraction and reflection also increases. - The intensity of the reflective ray increases, and the intensity of the refractive ray decreases - When angle of refraction = 90°, angle of incidence = critical angle
	- When angle of incidence > critical angle, total internal reflection occurs. - All light is reflected, none is refracted

Critical angle: the angle of incidence in denser medium, when angle of refraction is 90°

Conditions for Total internal reflection:

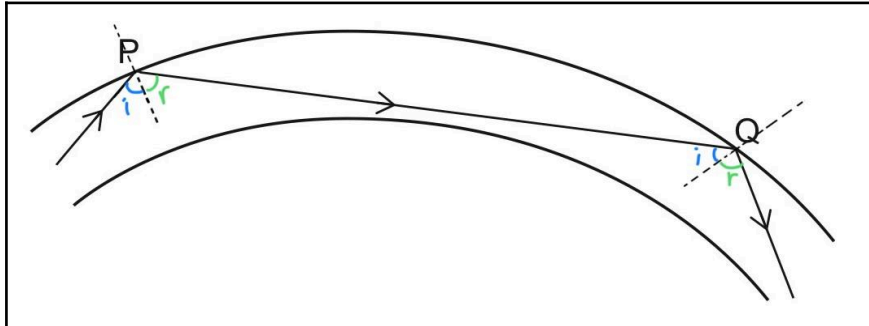
- Light ray travels from more dense to less dense medium
- Angle of incidence > critical angle
- No refraction of light
- Reflection of light - Angle of incidence is equal to angle of reflection

Finding refractive index using critical angle:

$$n = \frac{1}{\sin C}$$

Uses of total internal reflection:

Optical fibres:



Advantages of using optical fibres:

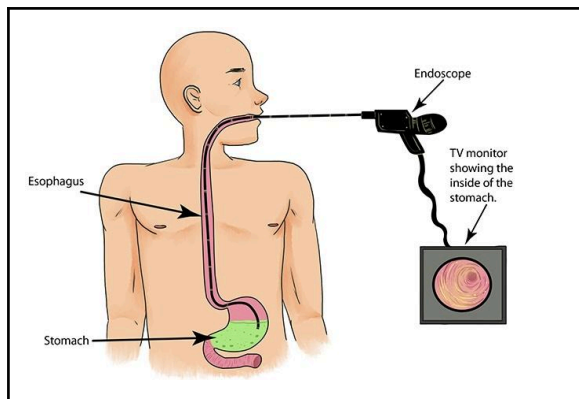
- Fast data transmission
 - Can carry large amounts of data
 - There is little data loss
-

In Communication:

- Light carries information (data) along optical fibres

In Medical imaging:

- Used in endoscope to see inside the body of patients



- Light ray from outside is transmitted through optical fibres, and undergo total internal reflection to prevent them from leaving optical fibres
- Light ray is reflected at inside of the body, and returns along optical fibres
- This light is sent to the screen, where it is displayed

Lenses

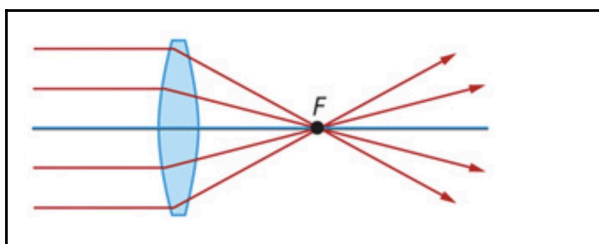
Principal axis: the line passing through the centre of the lens, perpendicular to its surface

Focal length: the distance between centre of lens and principal focus (focal point)

Principal focus (focal point): the point at which rays of light parallel to the principal axis converge after passing through a converging lens

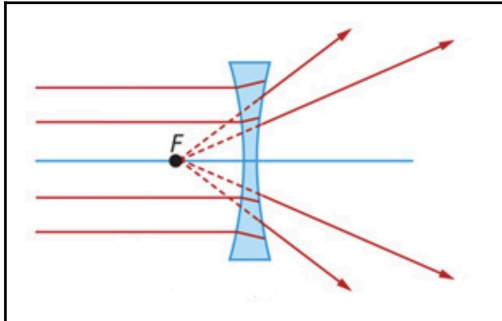
Converging lens:

- Parallel light rays are brought to a focus



Diverging lens:

- Parallel light rays spread out from a point



Images produced by lenses - real or virtual

- real: can be formed on a screen
 - virtual: cannot be projected onto a screen
-

Images from converging lens - ray diagrams

- Draw 3 rays (remember to include arrows)
 - One straight than to focal point
 - One directly through centre of lens
 - One to do all point than straight

Object beyond $2F$:

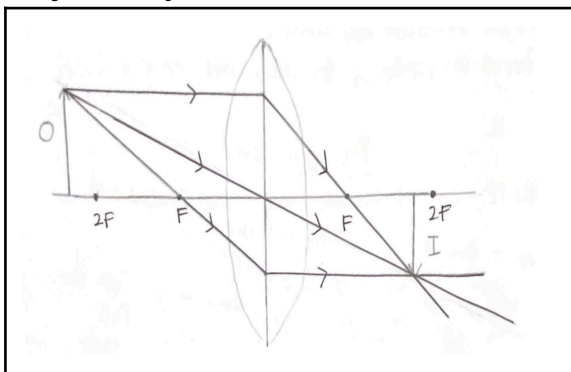


Image characteristics:

- Real
- Diminished
- Inverted
- Image between F and $2F$

Object is at $2F$:

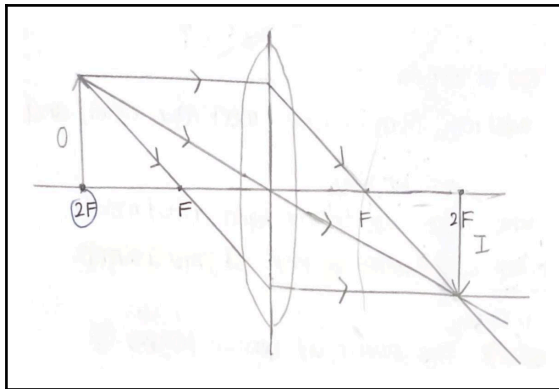


Image characteristics:

- Real
- Same size as object
- Inverted
- Image at $2F$

Object between F and $2F$:

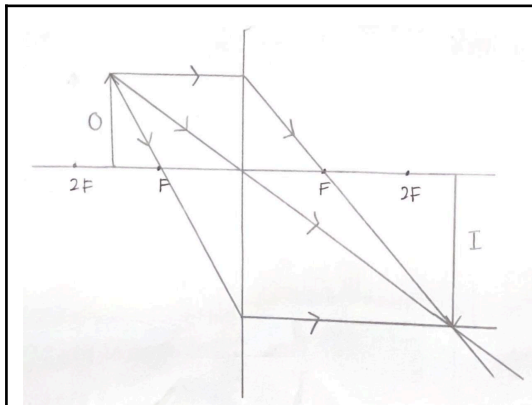
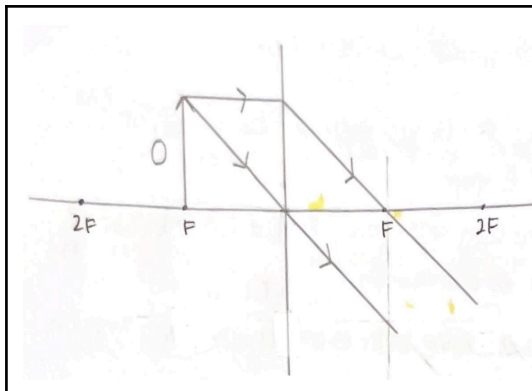


Image characteristics:

- Real
- Enlarged
- Inverted
- Image Beyond $2F$

Object is at F :



- Object cannot be focused because rays are parallel, they do not intersect

Object between F and lens:

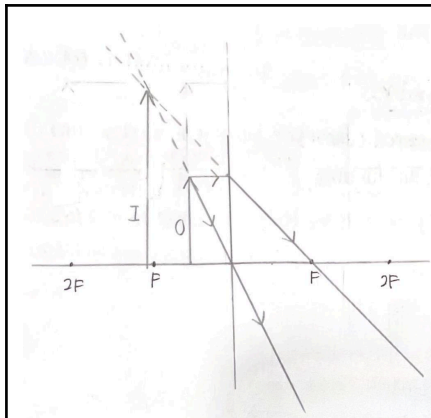


Image characteristics:

- Virtual
 - Enlarged
 - Upright
- Image is formed on the same side as the object
 - This case is the use of a single lens of a magnifying glass

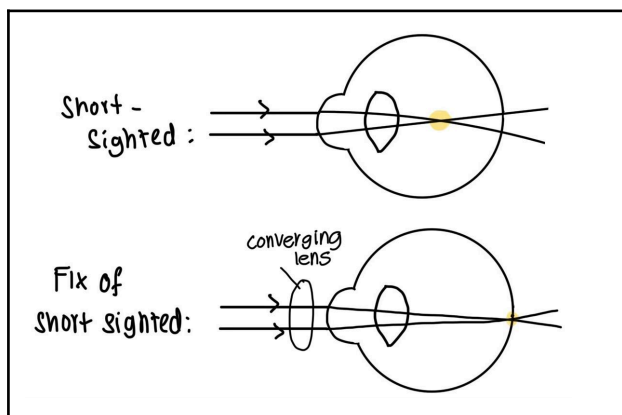
Overall trends:

- At range F - beyond $2F$, when the object moves towards the lens, the image moves away from the lens
 - For between F and lens, when object gets close to lens, image also gets closer, and image size will decrease
-

Correcting sight

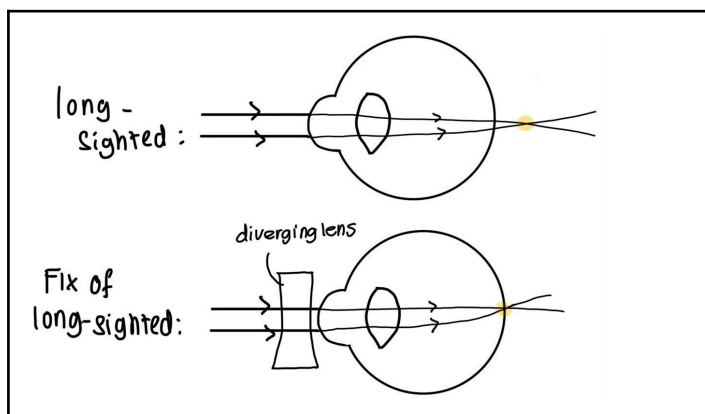
For short-sightedness:

- People who are short-sighted cannot see things that are far away, and can only see things that are close to them
 - This is because the eye refracts light too much, and brings it to a focus before it reaches the retina.
- This can be corrected by using a diverging lens, to refract light less, and bring it onto the retina



For long-sightedness:

- People who are long sighted cannot see things that are close to them, and can only see things that are far away
 - This is because the eye refracts the light too little, and brings it to a focus beyond the retina
- this can be corrected by using a converging lens, to refract light more and bring it onto the retina.



Dispersion of light

- Dispersion of light is the spreading out of white light into the colours of the spectrum
 - This is possible by passing it through a prism
 - White light (visible light) is a mixture of all the colours of the spectrum
 - Each colour has a different wavelength and frequency

Visible spectrum of light

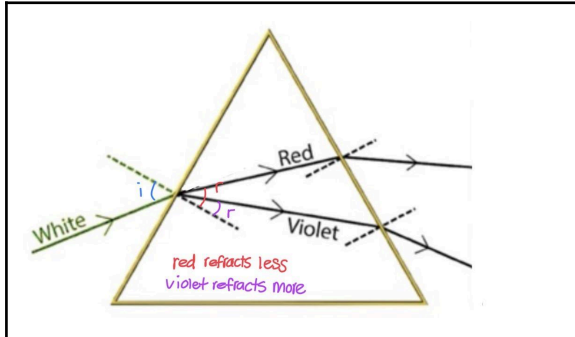
Visible light: the range of wavelengths which are visible to humans

Red -> orange -> yellow -> green -> blue -> Indigo -> violet

- Red has the longest wavelength, lowest frequency and lowest energy
- Violet has the shortest wavelength, shortest frequency and highest energy

Demonstrating dispersion of light

- When travelling from air to glass prism, all light rays will bend towards the normal as the speed will be slower for all of them



For Red light:

- Red light has the longest wavelength and fastest speed
- So, it refracts the least
- This also means that the refractive index of glass is lower for violet light

For Violet light:

- Violet light has the shortest wavelength and slowest speed
 - So, it refracts the most
 - This also means that the refractive index of glass is higher for violet light
-
- After the light rays exit the glass, all the different colours of light speed up and then travel at the original speed
 - This is 3×10^8 m/s
-

3.3 *Electromagnetic spectrum*

Electromagnetic spectrum

- They are all transverse waves
- They can travel through a vacuum
- They travel at the same speed in vacuum: 3×10^8 m/s

Radiowaves -> Microwaves -> Infrared radiation -> visible light -> Ultraviolet -> X-rays -> Gamma rays
Lowest energy _____> Highest energy
Lowest frequency _____> Highest frequency
Longest wavelength _____> Shortest wavelength

- The higher the frequency, the higher the energy of the electromagnetic wave

Electromagnetic waves with higher energy:

- Ultraviolet, X-rays, and Gamma rays
- They are highly ionising
- They are harmful to cells and tissues, causing cancer

Electromagnetic waves with lower energy:

- Radiowaves, Microwaves, Infrared radiation, Visible light
- They are less ionising
- They are less harmful to humans
- Useful for communication

*Note - Ionising means to remove electrons from atoms or molecules

Uses of electromagnetic waves

Radiowaves:

- Communication
- Radio frequency identification (RFID)
- Bluetooth
- Astronomy, radio telescopes

Microwaves:

- Communication
- Cooking in microwave oven

Infrared radiation:

- Thermal imaging in infrared cameras
- Intruder alarms and motion sensors
- Optical fibres to transmit data in communication and medical imaging
- Electric grills
- Remote controls

Visible light:

- Vision and photography
- Optical fibres to transmit data in communication and medical imaging

Ultraviolet:

- Fluorescence in security marking and detecting fake bank notes
- To sterilise surfaces and kill bacteria
- Tanning

X-rays:

- Medical imaging
- Security scanning

Gamma rays:

- To detect and treat cancer cells
- To sterilise food and medical equipment

Dangers of electromagnetic waves

Radiowaves:

- No known danger

Microwaves:

- Possible heat damage to internal organs

Infrared radiation:

- Heating effect of tissues, skin burns

Visible light:

- Very bright light can damage eyes

Ultraviolet:

- Eye damage when there are high levels of Ultraviolet radiation
- Skin cancer

X rays:

- Kills cells
- Mutation
- Cancer

Gamma rays:

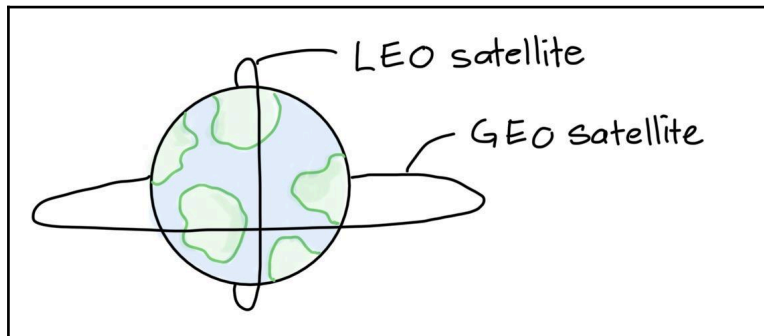
- Kills cells
- Mutation
- Cancer

To protect yourself from harmful effects:

- Limit your exposure to it
- Stay away from sources
- Wear protective clothing

Communication with artificial satellites

- Both uses microwaves



Geostationary satellites:

- Orbits at Earth's equator
 - Uses: Radio, Telecommunication broadcasting
- Can transmit larger volumes of data
 - Higher latency (delay)

Low Earth orbit satellites:

- Orbits at Earth's north and south poles
 - Uses: Monitoring weather, Military applications
- Does not transmit as much data as GEO satellite
 - Lower latency (delay)
-

System of communications

Bluetooth:

- Uses radio waves
- Because, radio waves pass through walls, but the signals are weakened on doing so

Mobile phones and wireless internet:

- Uses microwaves
- Because, microwaves can penetrate some walls, and only require a short aerial for data transmission

Optical fibres:

- Used for cable television and high-speed broadband
 - Uses infrared radiation and visible light
 - Because, glass is transparent to visible light and some infrared radiation
 - Visible light and short wavelength infrared can carry high rates of data
-

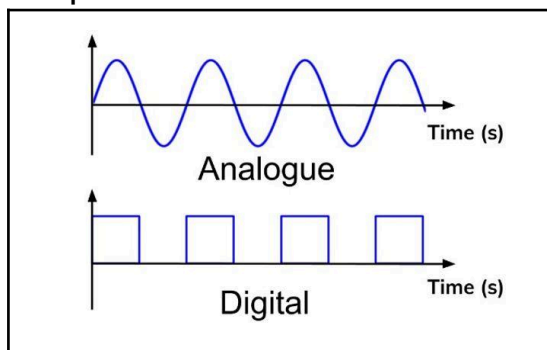
Digital and Analogue signals

- Analogue signals vary continuously: they can take any value
- Digital signals can only take one of 2 values (discrete)

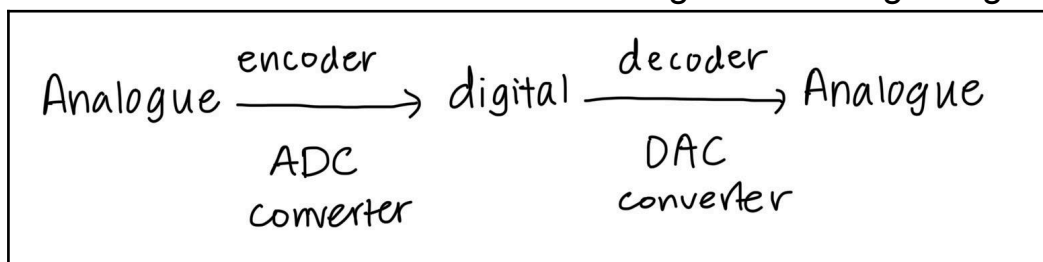
Advantages of digital signal:

- Can cover large distances
- Fast data transmission
- Signal can be regenerated, so there is minimal noise
- Extra data can be added so that signal can be checked for errors

Graphs:



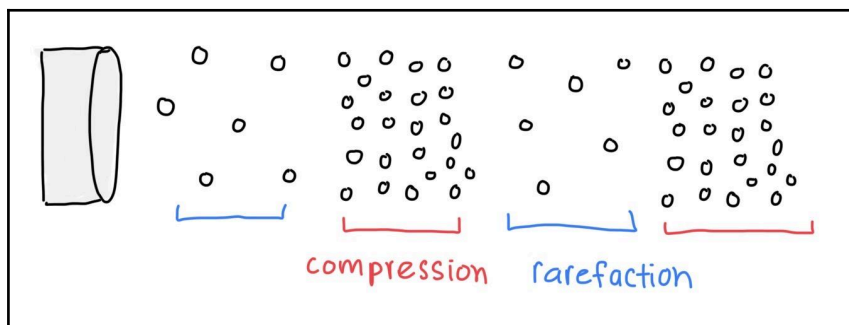
Sound waves can be transmitted as a digital or analogue signal:



3.4 Sound waves

Sound waves:

- Longitudinal waves
 - Require medium; sound cannot travel through a vacuum
 - Produced by vibrating sources -> when vibrating sources move back and forth, it creates pressure variations in the surrounding medium
-
- Sound travels fastest in solids, than in liquids, than in gases
 - Speed of sound in air = 330m/s - 350m/s
 - At 25°C, speed of sound in air is 330m/s
 - Speed of sound in hot air is faster than in cold air

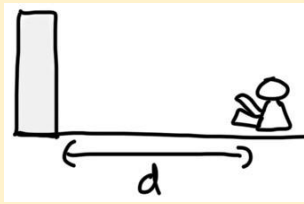


- The harder you bang, the larger the vibrations, and the louder the sound (larger amplitude)
- This causes air molecules to be closer together in compressions, and further apart in rarefactions

Echo: the reflection of sound waves

*be aware of distance -> whether need to double

To determine speed of sound in air - echo method at wall



- Stand 100m from a wall, this distance is measured using measuring tape.
- Clap wooden blocks, and immediately start stopwatch.
- Record time taken until the echo is heard. Record as t .
- Calculate speed ($s = 200 / t$)
- Repeat 2 more times and calculate average value

*note -> average value has to be calculated from the dependent variable (time)

Diffraction of sound waves:

- Sound can diffract around the corner of a building or through a doorway -> this is because the wavelength of sound is similar or larger than the size of the door gap / corner of building

	- When a person is walking on the other side of a building, she can hear the sound from the radio because the sound diffracts around the corner of the building
	- When a person is outside of a room, and the door is open, he can hear the sound from coming inside the room, because the sound diffracts through the doorway

Pitch and loudness of sound:

- Low pitch -> low frequency, long wavelength
 - High pitch -> high frequency, short wavelength
 - High volume -> large amplitude
 - Small amplitude -> low volume
-

Ultrasound: sound with a frequency greater than 20,000Hz

- Human audible range: 20 - 20,000Hz

Uses of ultrasound

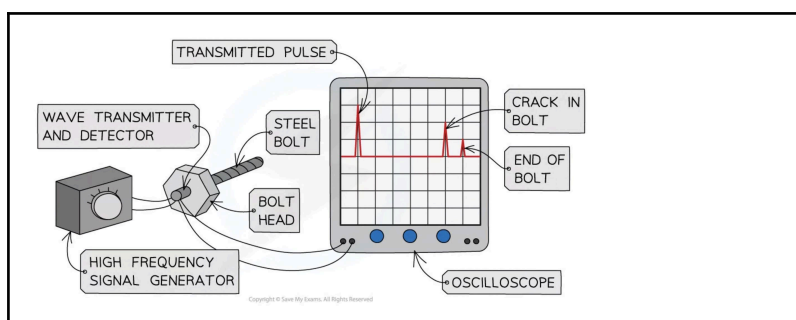
- Linked to echo formula -> as it is reflection of the ultrasound waves

Medical scanning of soft tissue:

- To construct Images of fetus in the womb
- To generate 2D images of organs and other internal structures

Non-destructive testing of materials:

- To check for cracks inside metal objects
- To generate images beneath surfaces

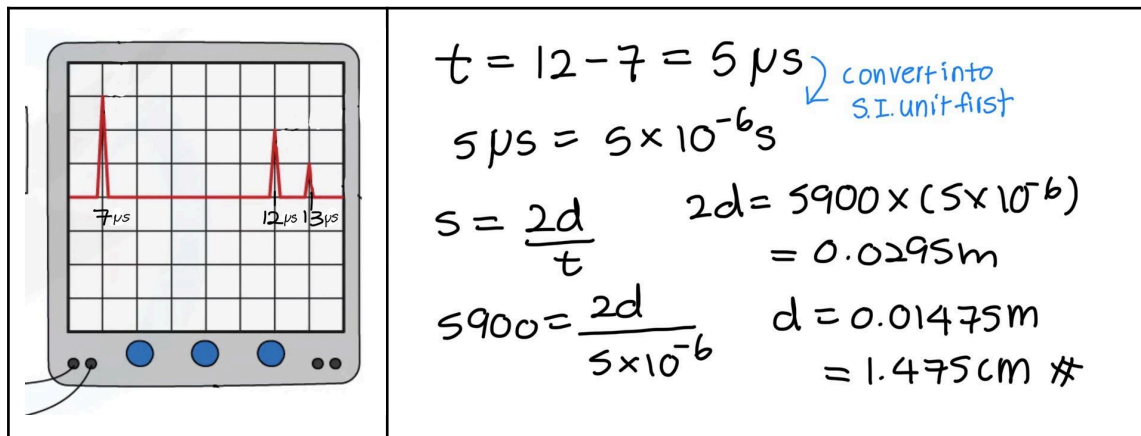


- Ultrasound waves that reach a crack in an object are reflected back from the crack earlier than the waves that are absorbed through the object
- The reflected waves are displayed as pulses on an oscilloscope

- So, for an object with no cracks, there should only be 2 pulses -> 1 for start of object and 1 for end of object

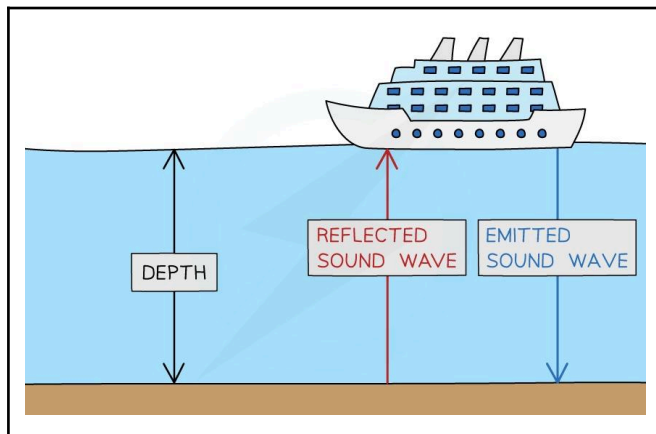
Example - calculation

- Provided: Speed of sound in the object is 5900m/s



Sonar:

- sound navigation and ranging for boats and some animals



- A sound wave is transmitted from the surface of the water
- The sound wave is reflected off the bottom of the ocean
- The time taken for the sound wave to return is recorded.
- This is used to measure the depth of the water

-> This can also be used to detect objects underwater